UNIT QUESTIONS

SKILLS PROBLEM SOLVING 7 **1**

a A rectangular block of metal of density 8000 kg/m^3 measures 2 cm by 3 cm by 10 cm. What is the mass of the block?

A 48 g
B 0.48 kg
C 4.8 kg
D 48 kg **(1)**

b Which of the following will exert the greatest pressure when standing on the ground?

A an elephant: weight 50 000 N, area of one foot 0.2 m^2
B atmospheric pressure: 10^5 Pa
C ballerina: weight 450 N, on one point, area 0.0015 m^2
D man: weight 800 N, area of one foot 0.025 m^2 **(1)**

SKILLS CRITICAL THINKING 5

c Which of the following descriptions most accurately describes a gas?

A very low density, easily compressed, cannot flow
B high density, cannot be compressed, definite shape
C very low density, easily compressed, expands to fill available space
D high density, difficult to compress, can flow **(1)**

d The specific heat capacity of a liquid is

A the amount of heat energy required to make 1 kilogram of the liquid boil
B the amount of heat energy required to raise the temperature of 1 kilogram of the liquid by 1 °F
C the amount of heat energy required to raise the temperature of 1 kilogram of the liquid by 1 °C
D the amount of heat energy required to raise the temperature of 1000 cm^3 of the liquid by 1 °C **(1)**

e Boyle's law states that

A if you double the pressure on a gas its volume doubles
B if you double the pressure on a gas its temperature halves
C if you double the pressure on a gas its volume doubles provided its temperature is constant
D if you double the pressure on a gas its volume halves provided its temperature is constant **(1)**

(Total for Question 1 = 5 marks)

SKILLS INTERPRETATION 7 **2**

Look at the diagram below showing a small gas bubble in a glass of fizzy drink, like cola.



a Copy the diagram and use eight arrows to shows how the pressure of the liquid acts on the gas bubble. **(2)**

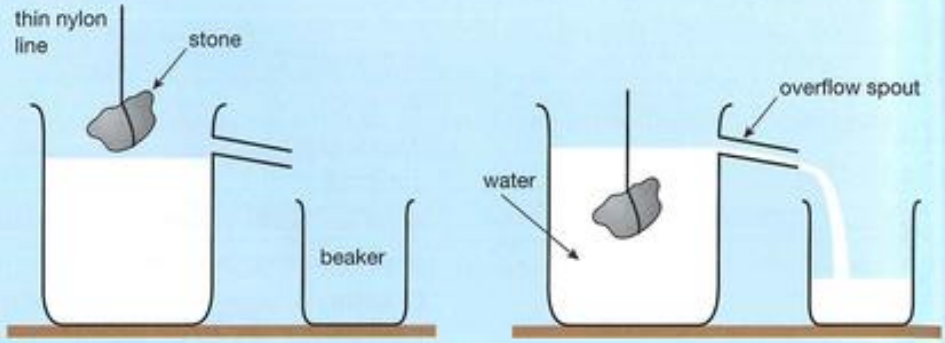
SKILLS REASONING (7)

b Describe what happens to the bubble as it floats up towards the top of the drink and explain the reason for your answer (assume that the amount of gas in the bubble does not change as the bubble rises up through the cola). **(3)**

(Total for Question 2 = 5 marks)

SKILLS EXECUTIVE FUNCTION

3 The diagram shows part of the apparatus used by a student to find the density of a stone.



(4) **a i** What is the student measuring? **(1)**

(3) **ii** What other piece of apparatus is required to complete the measurement? **(1)**

iii What precautions should the student take to ensure that the measurement is accurate? **(1)**

SKILLS CRITICAL THINKING (5)

b i State the equation that the student needs to calculate the density of the stone. **(1)**

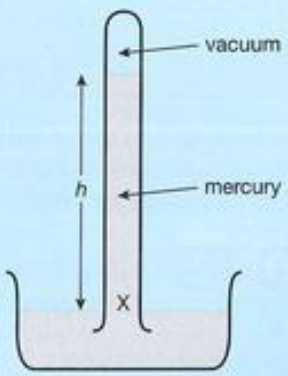
SKILLS EXECUTIVE FUNCTION

ii What other measurement must be made? State the name of the apparatus needed to make this measurement. **(1)**

(Total for Question 3 = 5 marks)

SKILLS PROBLEM SOLVING (7)

4 The diagram shows a simple mercury barometer made by filling a strong glass tube with mercury and turning it upside down (with the open end of the tube closed) in a dish of mercury and then opening the end below the level of mercury in the dish. Mercury runs out of the tube until the pressure of the mercury at X is equal to the pressure of the air acting on the surface of the mercury in the dish.



The height, *h*, of mercury in the tube is a measure of air (atmospheric) pressure. A typical value for *h* is 76 cm.

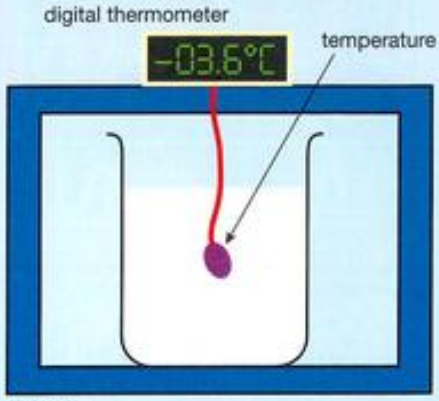
The density of mercury is 13 600 kg/m^3. Use this value to calculate the atmospheric pressure in Pa (pascals). **(3)**

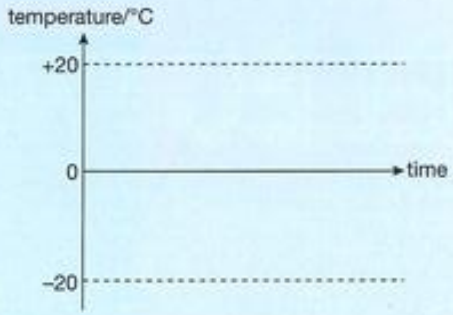
(Total for Question 4 = 3 marks)

PHYSICS ONLY

SKILLS INTERPRETATION 5

5 A beaker filled with water initially at 20 °C is cooled in a freezer until its temperature drops to −20 °C (the temperature inside the freezer). The temperature in the beaker is measured with a digital thermometer and recorded at regular intervals until it reaches −20 °C.





a Copy the graph axes shown and sketch how the temperature of the contents of the beaker changes with time from its starting temperature of +20 °C until it reaches −20 °C. **(5)**

SKILLS ANALYSIS

b Referring to your sketch graph explain what happens to the water as it is cooled from 20 °C to −20 °C. **(5)**

(Total for Question 5 = 10 marks)

SKILLS PROBLEM SOLVING **6** An electric kettle rated at 2.5 kW of mass 0.5 kg contains 1.5 kg of water at 20 °C. The kettle is turned on and the kettle and its contents are heated to 100 °C.

The s.h.c. (specific heat capacity) of the metal the kettle is made of is 500 J/kg °C and the s.h.c. of water is 4200 J/kg °C.

a Calculate the total amount of heat energy needed. **(5)**

b How long must the kettle be switched on for to heat the kettle and water to 100 °C? Give your answer in minutes to two significant figures. **(4)**

SKILLS REASONING

c In practice it will take longer than this to boil the water in the kettle. Explain why. **(1)**

HINT

You may need to refer to Chapter 2 to find the relationship between electrical power, energy and time.

(Total for Question 6 = 10 marks)

END OF PHYSICS ONLY